ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4 (Q2 struck out – not to be considered)

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 μm = 6.4 mm; FOV = $W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

(b) LSB = $5\text{ V}/2^{12} = 5/4096 = 1.2207\text{ mV} = 1.2207\times10^{-3}$ V.

ΔF = LSB/S with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207\times10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}\,x(kT) + \int_0^T e^{A(T-s)}\,B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}\,B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substitutting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$$\mathbf{x_{k+1} = 0.8187\,x_k + 0.0906\,u_k}$$

Question 2 [CROSSED OUT BY STUDENT]

~~*Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).*~~

~~**(a)** Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\,x_2 - mg\ell \sin x_1]$.~~

~~Equilibrium: $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Because $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.~~

~~**(b)** The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.~~

~~$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31\ (s^{-2})$. $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.~~

~~**$A = [0, 1; -61.31, -2.5]$**, and sence $\partial f_2/\partial\tau = 1/J = 50$, **$B = [0; 50]$**.~~

~~Chek: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).~~

~~**(c)** Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.~~

~~Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torqe would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torque would be non zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a steady error.~~

~~Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\,e = 2.4525\cos e$. Fixed point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.~~

~~e = 0.04899 rad × (180/π) = 2.807°, so θ_{ss} = 87.19°. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because K_p is much larger than mgℓ (K_p/mgℓ ≈ 20).~~

~~**(d)** Steady state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: steady-state error is zero**.~~

~~Poles: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell\cos\theta_d)\,\delta\theta = 0$, characteristic polynomal $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell\cos\theta_d$. After compensaton the added term mgℓ sin θ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.~~

~~The two differ only by mgℓ cos θ_d, and for θ_d = 90° this is 2.4525 × 0 = 0. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because cos 90° = 0; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3 × 0.02… i.e. it would move the poles.~~

~~*Remark: the sag of a PD loop under gravity shrinks as K_p is increased.*~~

~~*(Cancelled - please do not evaluate this question.)*~~

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is T = 1/f_s = 1 ms.

(a) Reason 1 – noice amplification: the backward diffrence $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly 1/T = 1000 s^{-1}, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder q = 2π/4096 = 1.53×10^{-3} rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20$ = 125.66 rad/s. The bilinear (Tustin) transform replaces s by (2/T)(z − 1)/(z + 1) with T = 1/f_s = 0.001 s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a)\,z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a)\,z^{-1}]$.

Dividing through by (1 + a): a/(1 + a) = 0.062832/1.062832 = 0.059117 and (1 − a)/(1 + a) = 0.937168/1.062832 = 0.881767. Hence with u = differenced signal and y = filtered velocity: **y_k = 0.05912 (u_k + u_{k-1}) + 0.88177 y_{k-1}**. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): y = 2(0.05912)u + 0.88177 y ⇒ y(1 − 0.88177) = 0.11824 u ⇒ y/u = 0.11824/0.11823 = 1.000. In general H(1) = 2a/[(1 + a) − (1 − a)] = 2a/2a = 1 exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: O = [C; CA] = [1, 0; 0, 1], full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and C = [1, 0], LC = [ℓ_1, 0; ℓ_2, 0], so A − LC = [$-\ell_1$, 1; $-\ell_2$, −2].

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2)$ = $s^2 + (2 + \ell_1)\,s + (2\ell_1 + \ell_2)$.

Both estimator poles at s = −10 means the desired polynomal is $(s + 10)^2 = s^2 + 20\,s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

L = [18, 64]^T. Verification: A − LC = [−18, 1; −64, −2] has trace −20 and determinant (−18)(−2) − (1)(−64) = 36 + 64 = 100, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) An observer works only when there is no noise, so it has no advantage; a filter is always better because it is simpler.

The filter is safer when the model is perfectly known.

Question 4

Given: plant $\theta\ddot{} + a\theta\dot{} + c\theta = b\,u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\,r - k_1\theta - k_2\theta\dot{}$.

(a) Closed loop: $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$.

$s = [-4 \pm \sqrt{(16 - 32)}]/2 = -4/2 \pm \sqrt{(-16)} = \mathbf{-2 \pm 4j}$.

DC gain $= N_{ff}/8 = 1$, so $\mathbf{N_{ff} = 8}$.

(b) With the actual b = 0.75 and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta\ddot{} + 3.5\theta\dot{} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stabel.

(c) Take the state $z = [\theta, \theta\dot{}, \xi]^T$. The plant gives $\theta\ddot{} = -a\theta\dot{} - c\theta + b\,u$ with $u = -k_1\theta - k_2\theta\dot{} + K_I\xi$, and $\xi\dot{} = r - \theta$. With r = 0 (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired pols −4 and −2 ± 2j: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by (s + 4): $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with a = 2, b = 1, c = 6: s^2: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. s^1: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. s^0: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and (8)(24) = 192 > 32, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\theta\dot{} = \theta\ddot{} = 0$ and, crucially, $\xi\dot{} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\xi\dot{} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\mathbf{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrum (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrator windup. If the acuator saturates (torqe limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the actuator is saturated, conditional integration, or back-calculation that feeds the diffrence between commanded and saturated torque back to the integrator.